Swiss Institute of Bioinformatics

PR STATS

Metagenomic Data Analysis of Microbial Communities (MGMC01)

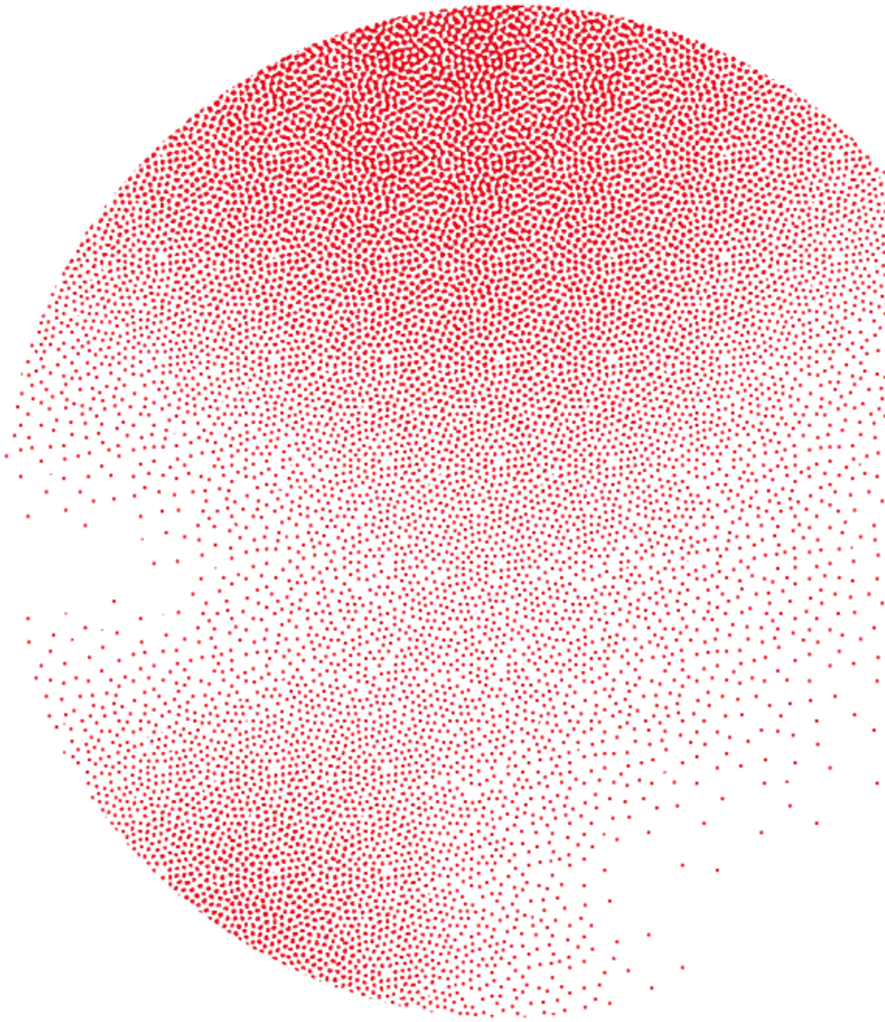
Day 4 - Session 2

PhD Jeferyd Yepes-García
University of Fribourg
October 29th, 2026

elixir SWITZERLAND

This presentation

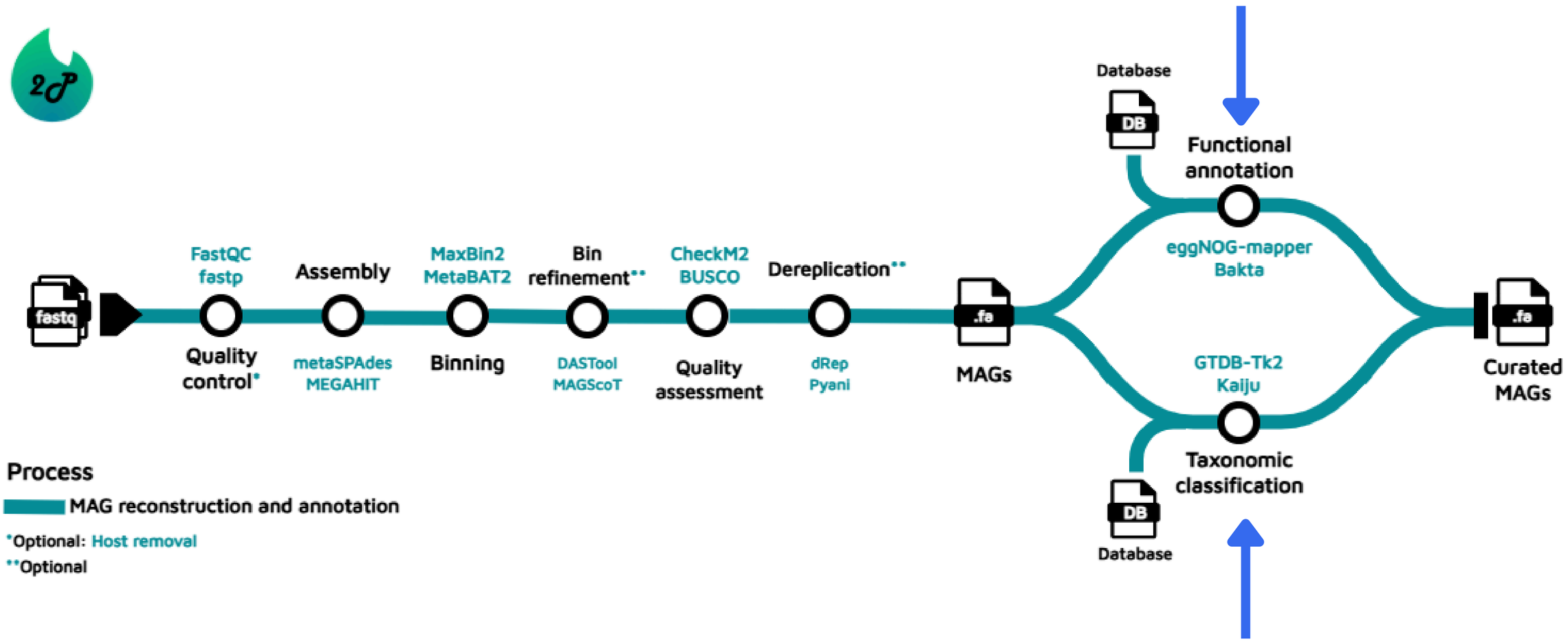
- Learning outcomes
- Taxonomic classification
- MAG de-replication
- Functional annotation
- Tools for functional annotation
- Pathway reconstruction
- Pangenome
- Visualization
- Summary



Learning outcomes

- Recognize both the goals and challenges of functional annotation of metagenome-assembled genomes (MAGs).
- Identify the most suitable tools to annotate MAGs according to the objectives of the study.
- Understand the purpose and scope of different tools designed to infer the metabolic potential of MAGs such as KEGG Decoder, COGclassifier, dbCAN3 (CAZymes) and antiSMASH (biosynthetic gene clusters), among others.

Once the MAGs are recovered



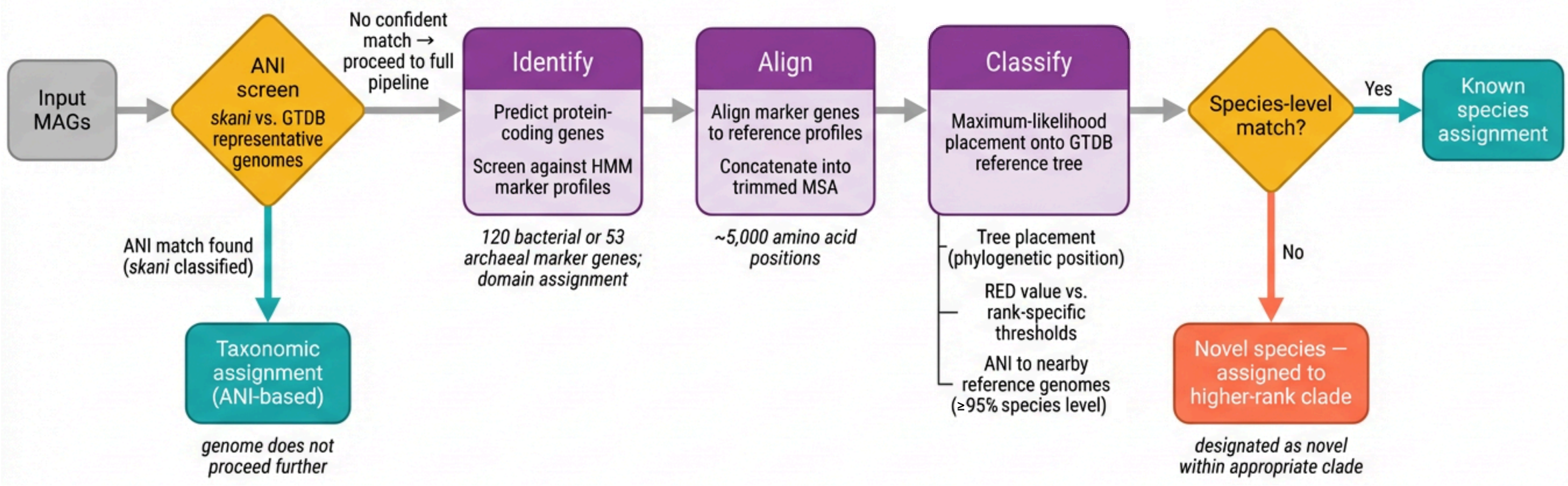
Taxonomic classification

GTDB: A standardized genome-based taxonomy for Bacteria and Archaea.



Taxonomic classification

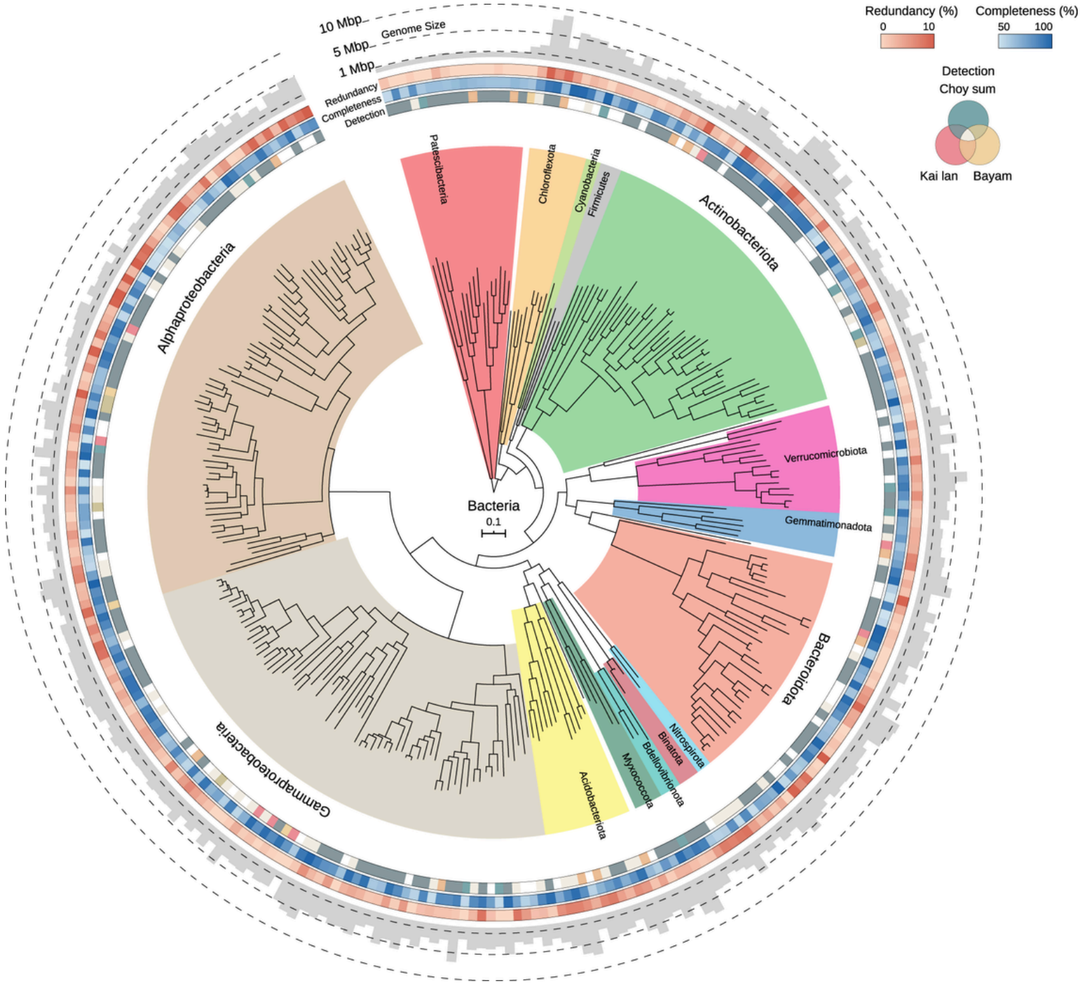
GTDB-Tk2: Assigns taxonomy to genomes based on GTDB using phylogenetic placement.



Alternatives: BAT, AutoMLST2, DFAST_QC.

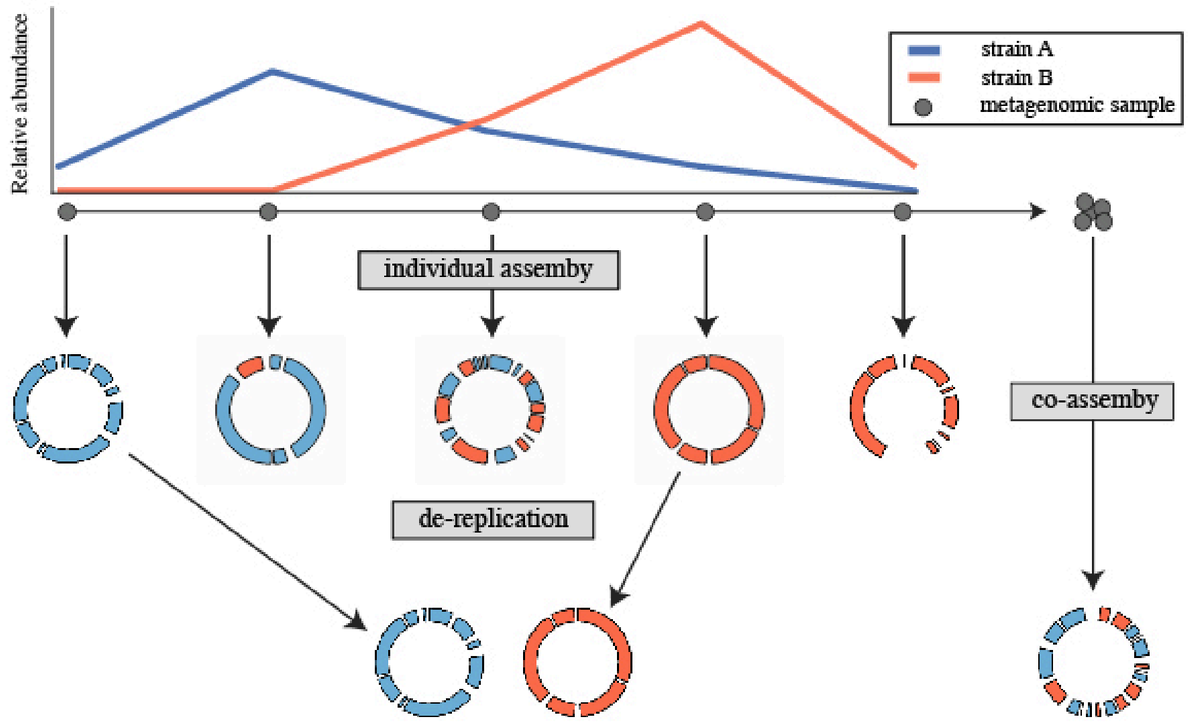
Taxonomic classification

GTDB-Tk2: Assigns taxonomy to genomes based on GTDB using phylogenetic placement.



MAG de-replication

Process of identifying sets of genomes that are the “same” in a list of genomes, and removing all but the “best” genome from each redundant set.



Tools:

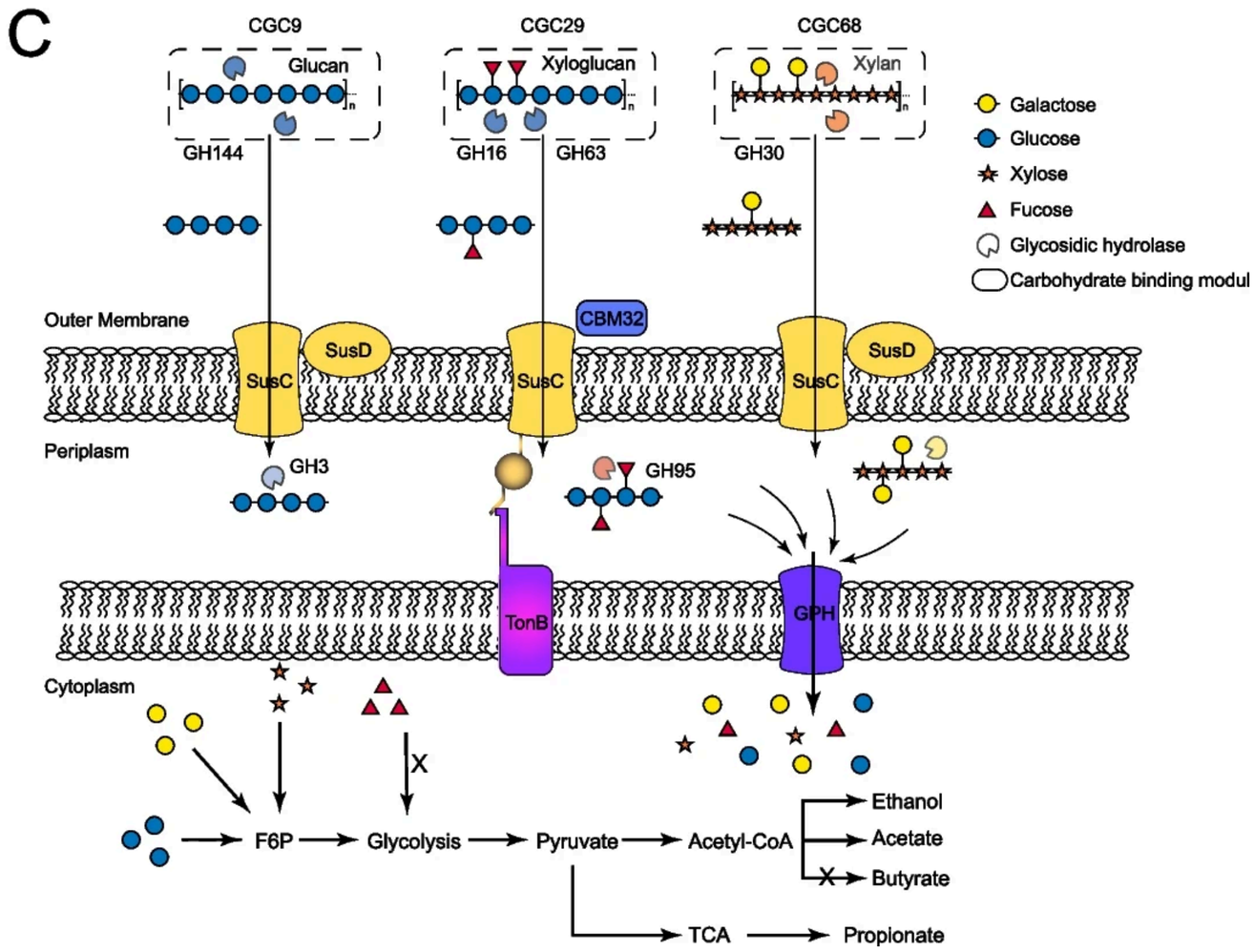
- dRep
- FastANI
- pyANI
- Sourmash

The removal of closely related genomes leads to losses of population-specific auxiliary genes.

Olm et al. (2017)

Functional annotation

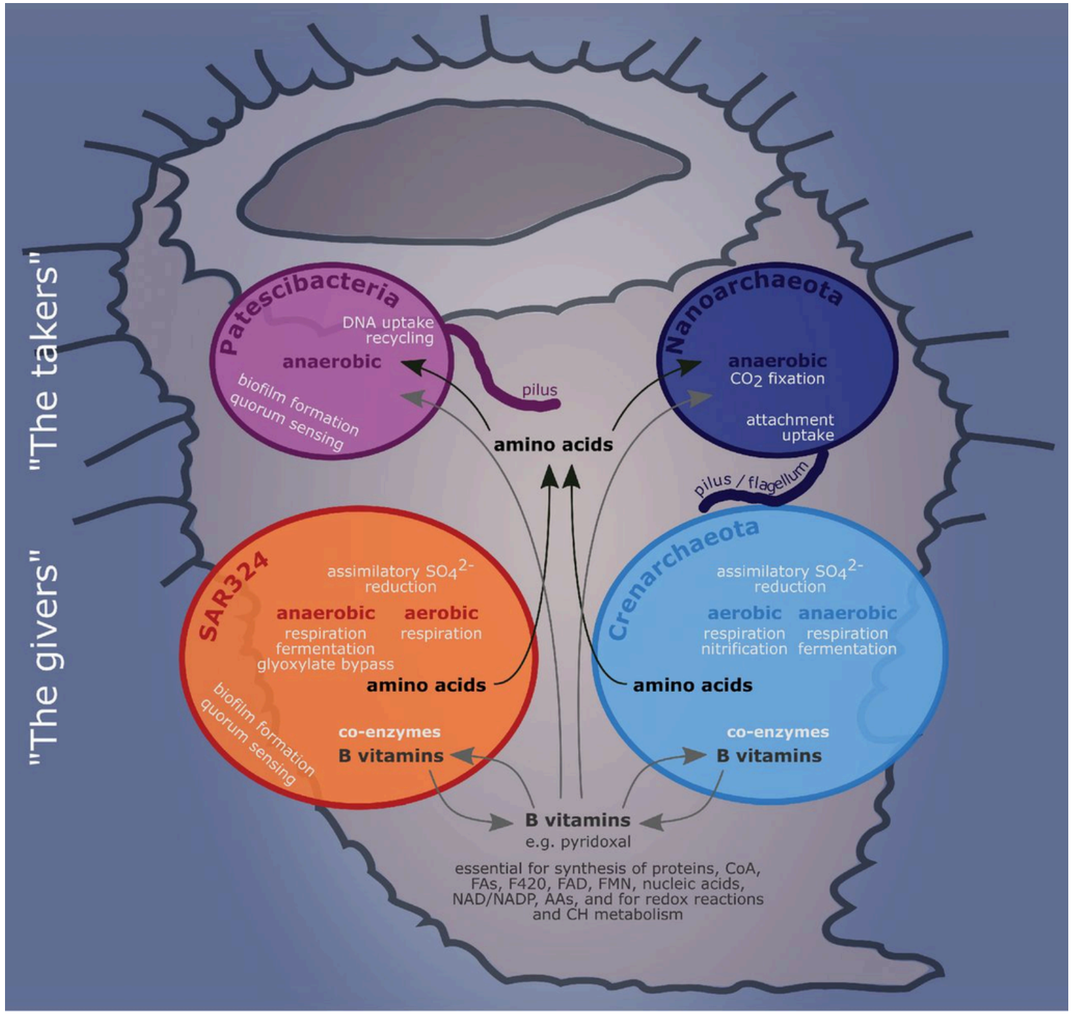
- Linking genetic content to ecological roles.
- Predicting metabolic pathways, environmental impacts, biotechnological potential.



Huang et al. (2023)

Functional annotation

- Metabolic interactions among taxa.
- Relationship and phenotypic associations with hosts.



Bayer et al. (2020)

Functional annotation

Gene Prediction

Prodigal
MetaGeneMark

Characterization of genomic features.

Functional Annotation

BLASTp
HMMER

Biological information.

Pathway Reconstruction

KEGG

Integrating gene functions into biological pathways.

Gene prediction

Identification of coding sequences (genes), typically detecting open reading frames (ORFs) and translating them into proteins.

Prokka

https://github.com/tseemann/prokka

Rapid annotation tool of bacterial genomes
Numerous outputs

Tool (reference)	Features predicted
Prodigal (Hyatt 2010)	Coding sequence (CDS)
RNAmmer (Lagesen *et al.*, 2007)	Ribosomal RNA genes (rRNA)
Aragorn (Laslett and Canback, 2004)	Transfer RNA genes
SignalP (Petersen *et al.*, 2011)	Signal leader peptides
Infernal (Kolbe and Eddy, 2011)	Non-coding RNA

Seemann, 2014

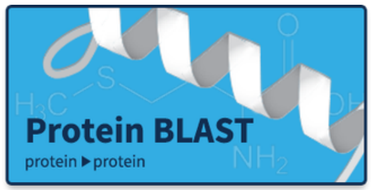
Alternatives:

- Bakta
- MicrobeAnnotator
- DFAST
- ePath

Biological functions

Assignment of biological functions to predicted genes or proteins by comparing them to known databases (e.g., pathways, domains, orthologs).

1. Homology-based methods:



diamond

2. Domain and Motif Detection:

HMMER

InterPro

Databases

EggNOG

KEGG Kyoto Encyclopedia of Genes and Genomes

Pfam

NIH National Library of Medicine
National Center for Biotechnology Information

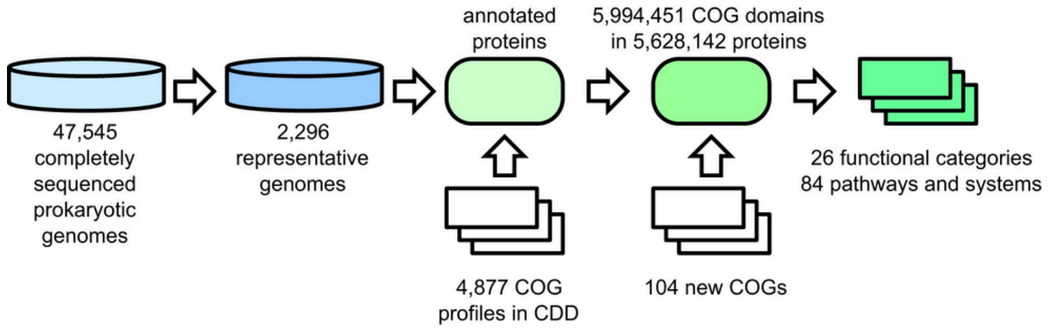
Database of Clusters of Orthologous Genes (COGs)

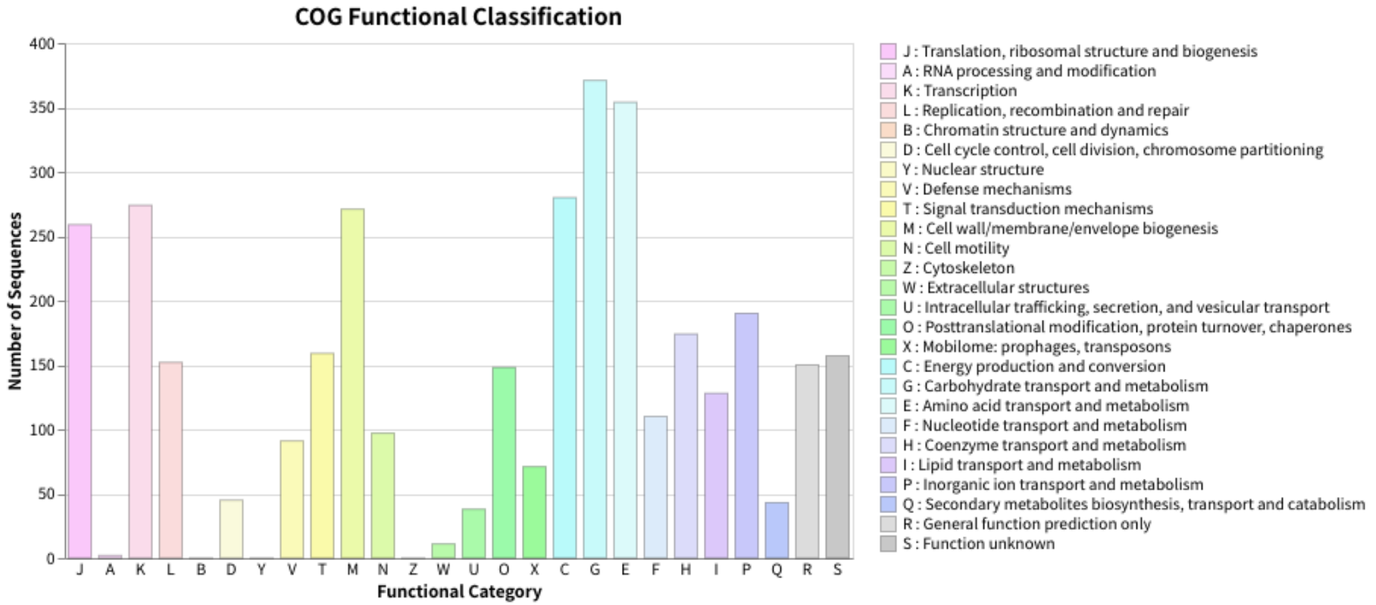
COG DATABASE | COG CATEGORIES | PATHWAYS | WEB SERVICES | COG PROJECT | CONTACT

SIB

Biological functions

COG database and COGClassifier: Assigns proteins or genes to COG functional categories by comparing sequences to the COG database.



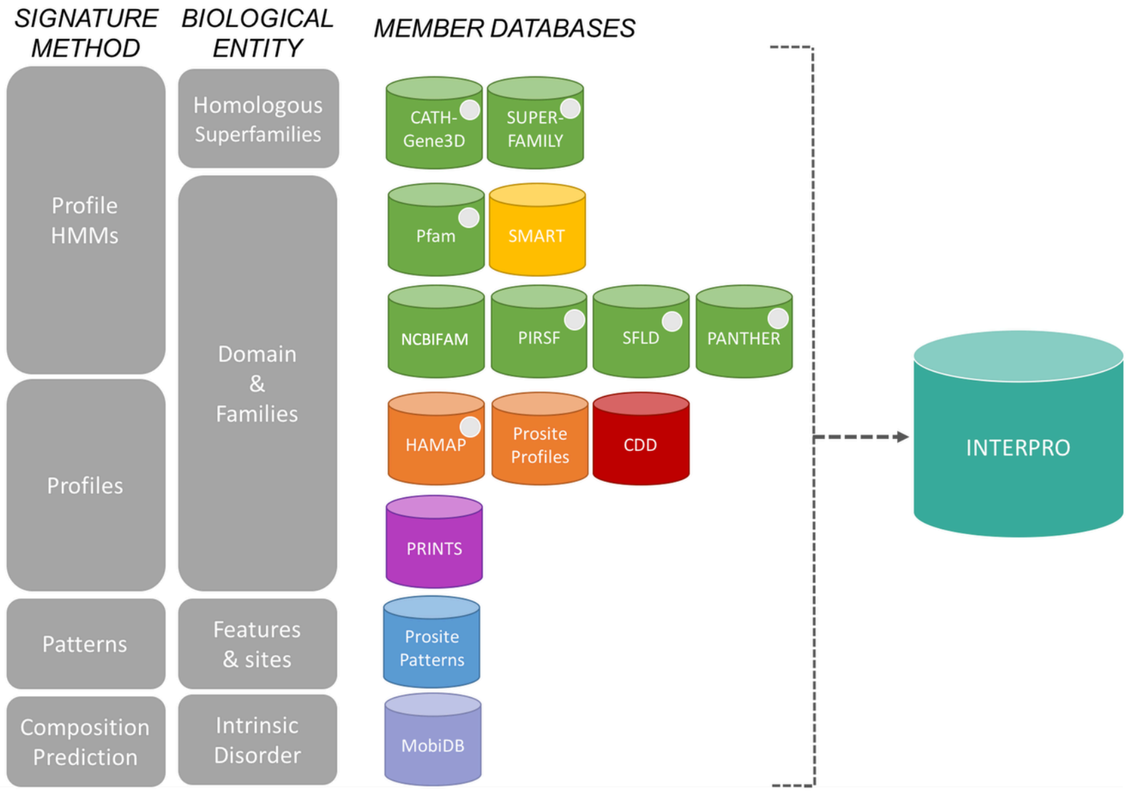


Method:

- RPS-BLAST

Biological functions

InterProScan: Integrates multiple protein signature databases to identify domains, motifs, and functional sites in predicted proteins.

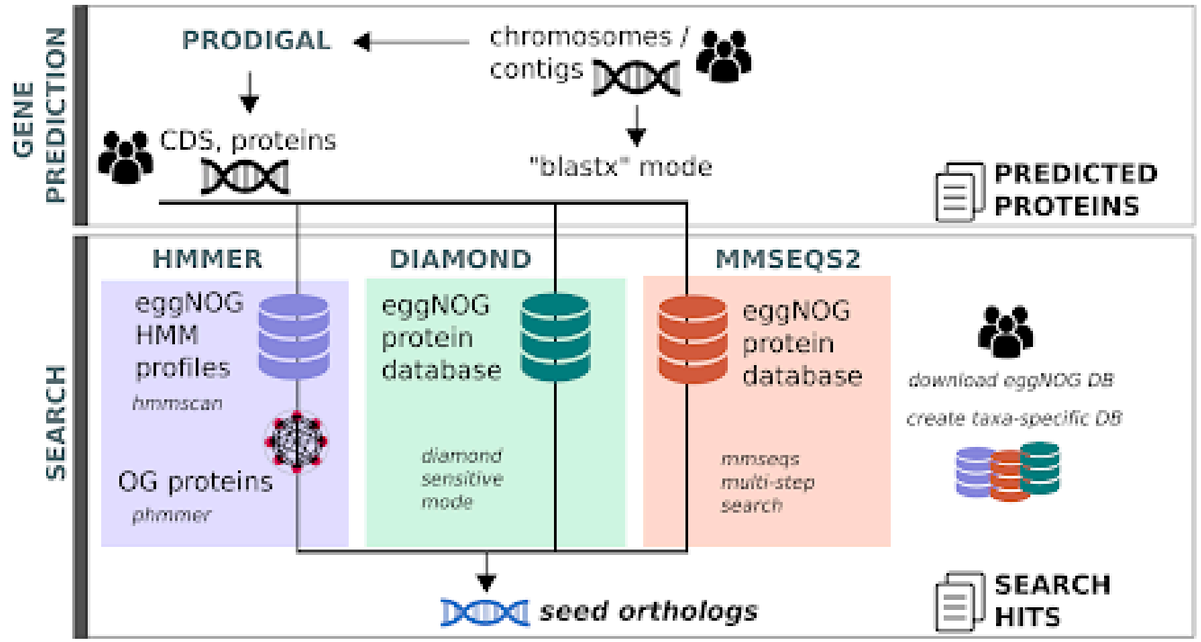


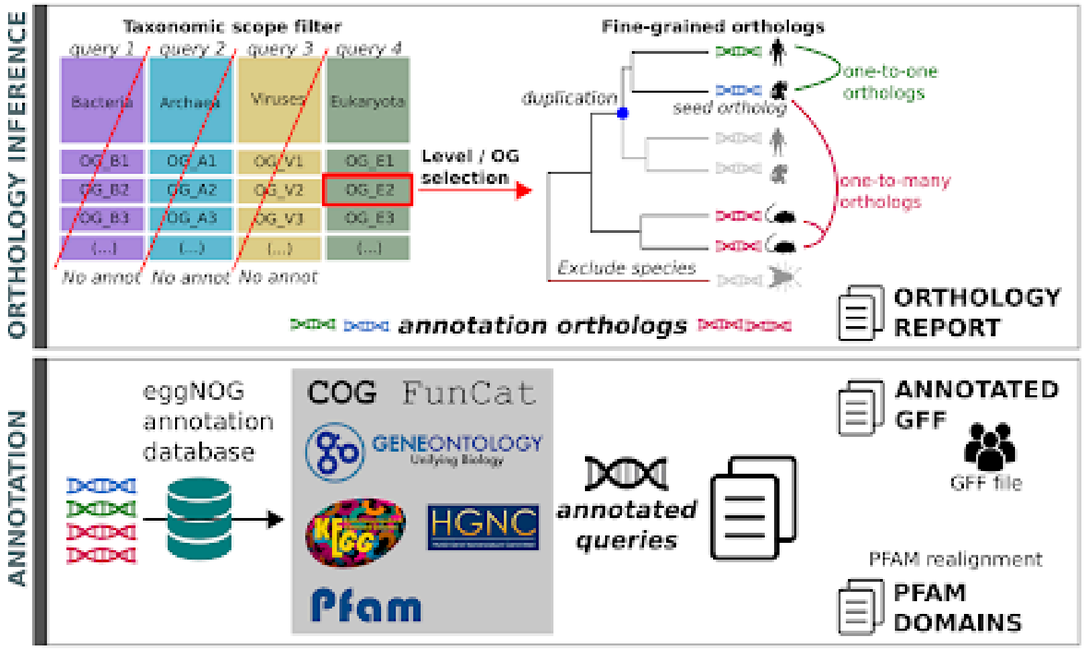
Methods:

- HMMER
- motif scanning

Biological functions

eggNOG mapper: Annotates genes based on orthology relationships, providing functional predictions, COG categories, and pathway information.



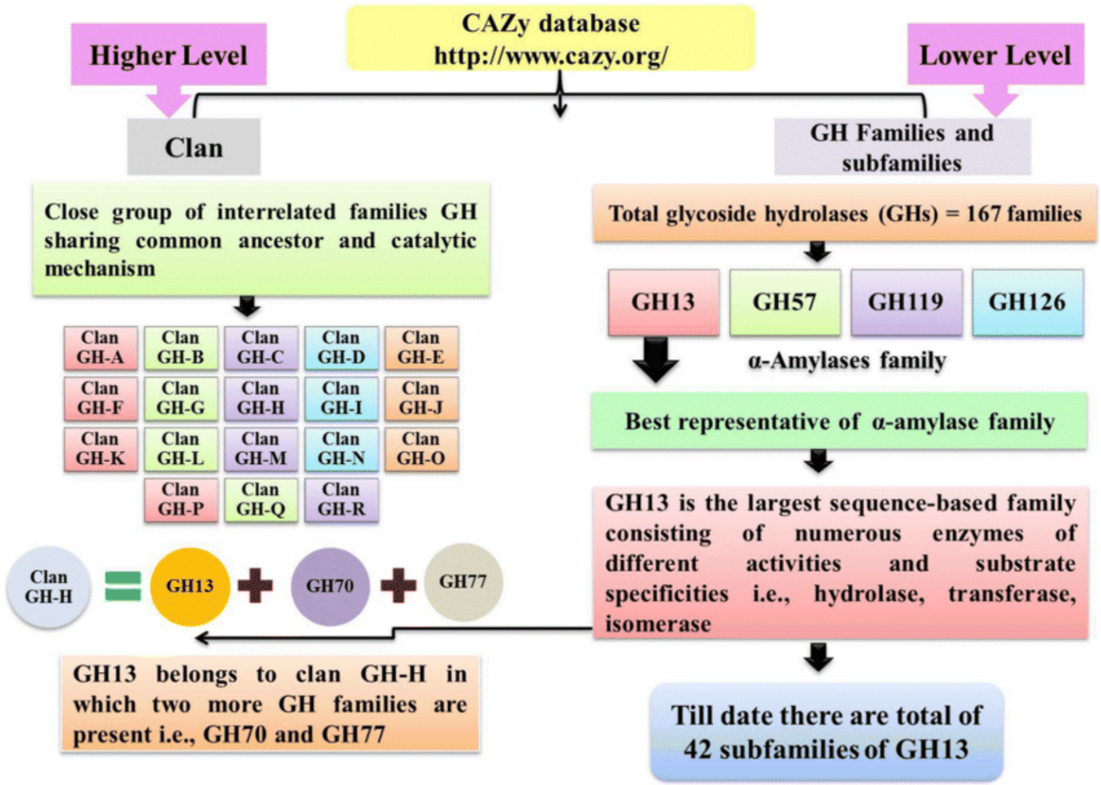


Alternatives:

- DeepNOG

Biological functions

dbCAN3: Identifies carbohydrate-active enzymes (CAZymes) involved in polysaccharide degradation and synthesis.





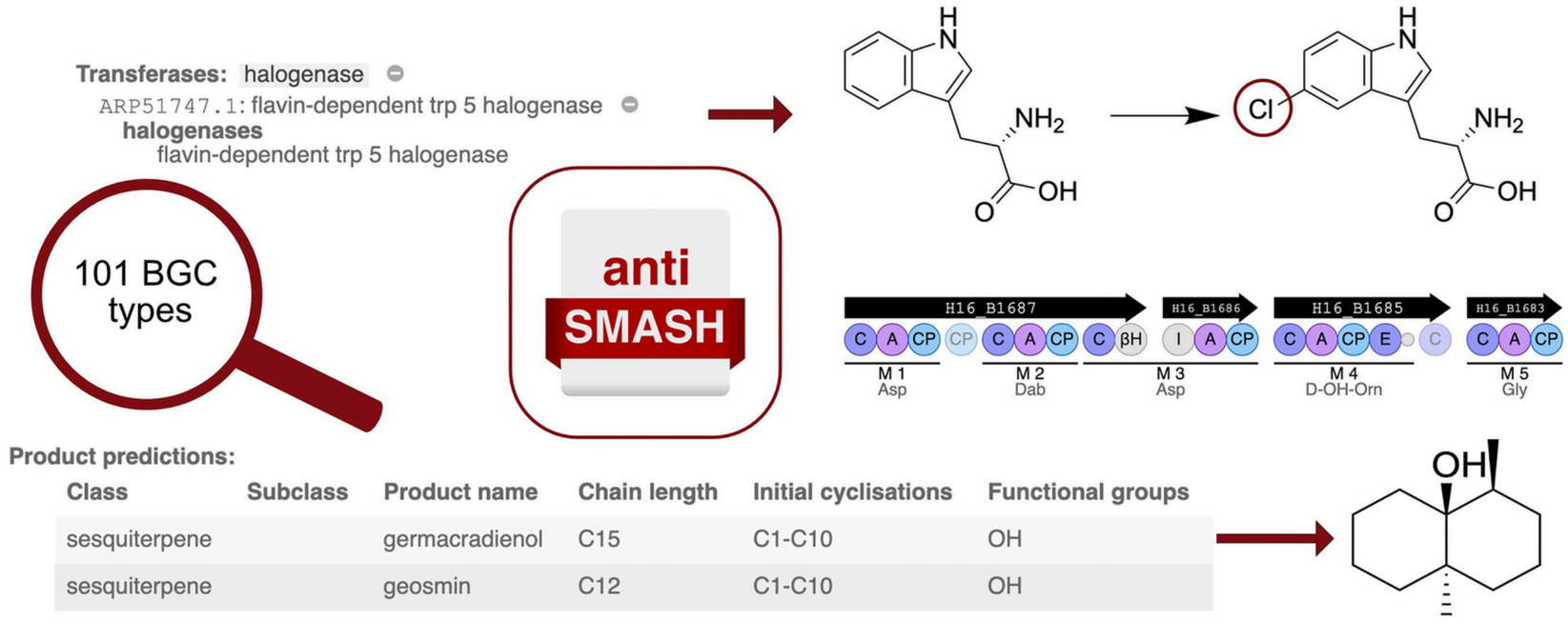
- DIAMOND (CAZy)
- HMMER (CAZy)
- CAZyme gene cluster (CGCFinder)
- Substrate prediction (dbCAN-sub)

Biological functions

antiSMASH: Detects and annotates biosynthetic gene clusters responsible for secondary metabolite production.

Two modes:

- Rule-dependent
- Rule-independent



Alternative:

- CluSeek

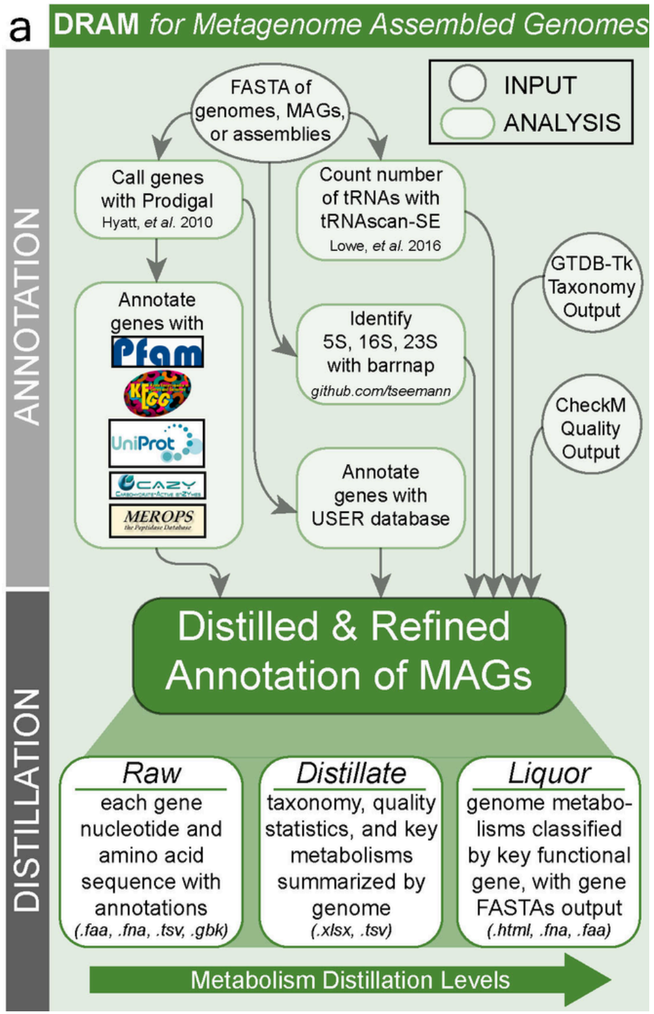
Biological functions

DRAM: Annotates microbial genomes and MAGs with a focus on metabolic functions and produces summarized metabolic profiles.

Default setup requires at least 500 GB of storage and 220 GB of RAM.

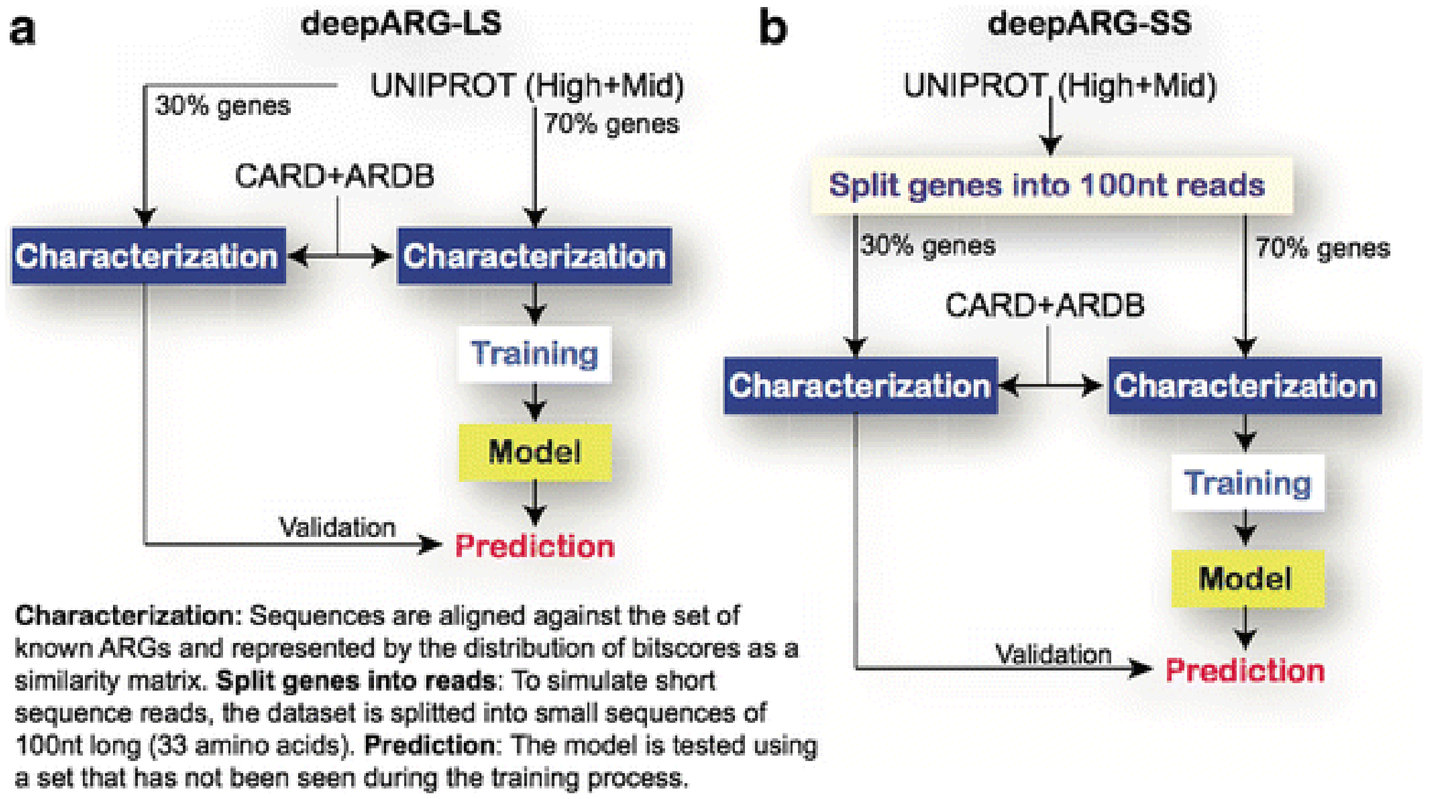
Alternatives:

- MetaSanity
- UPIMAPI



Biological functions

deepARG: Uses deep learning to identify and classify antibiotic resistance genes from sequence data.



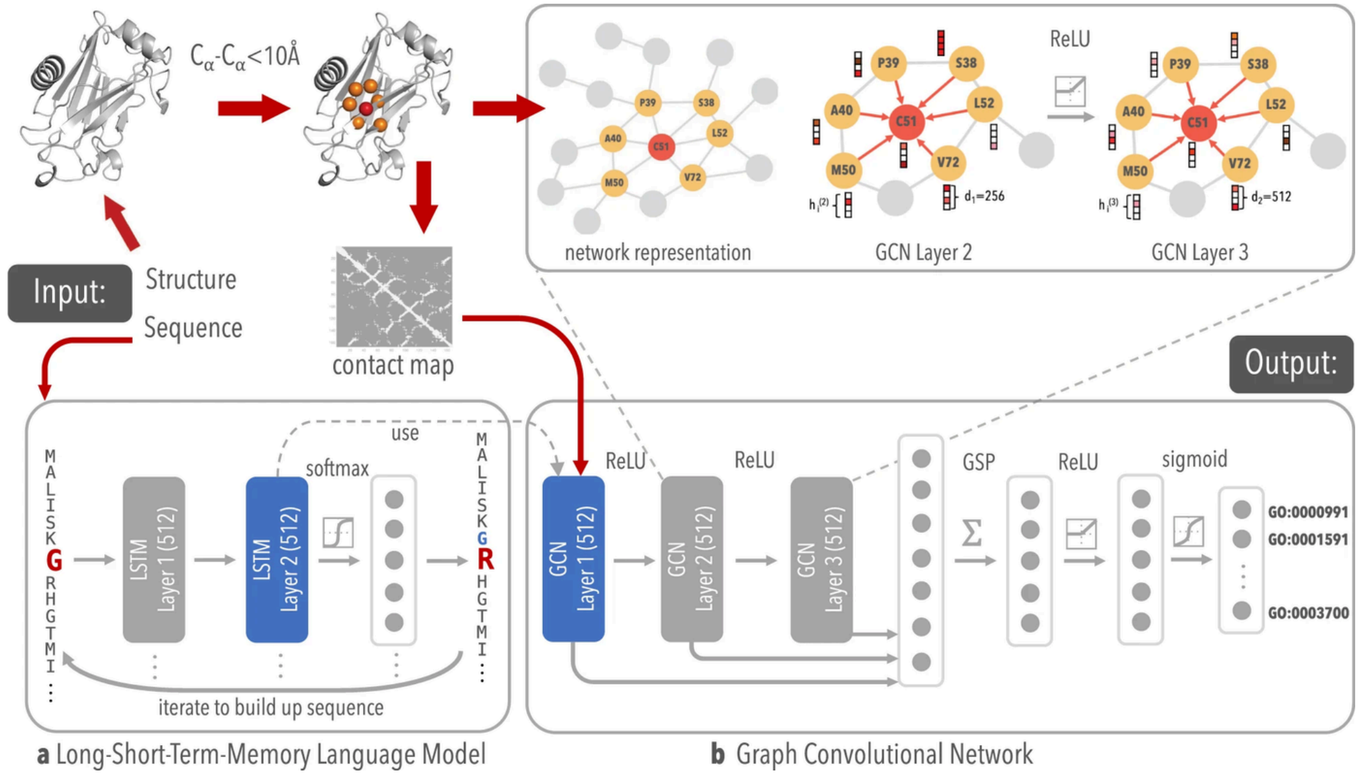
Characterization: Sequences are aligned against the set of known ARGs and represented by the distribution of bitscores as a similarity matrix. **Split genes into reads**: To simulate short sequence reads, the dataset is splitted into small sequences of 100nt long (33 amino acids). **Prediction**: The model is tested using a set that has not been seen during the training process.

Alternatives:

- AMRFinderPlus
- ResFinder/RGI

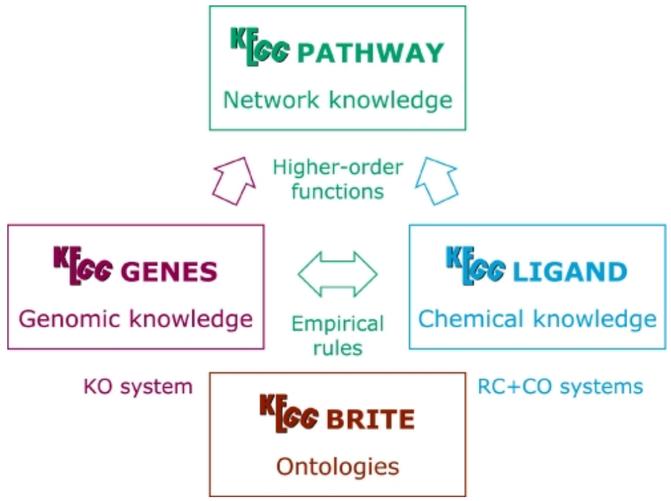
Biological functions

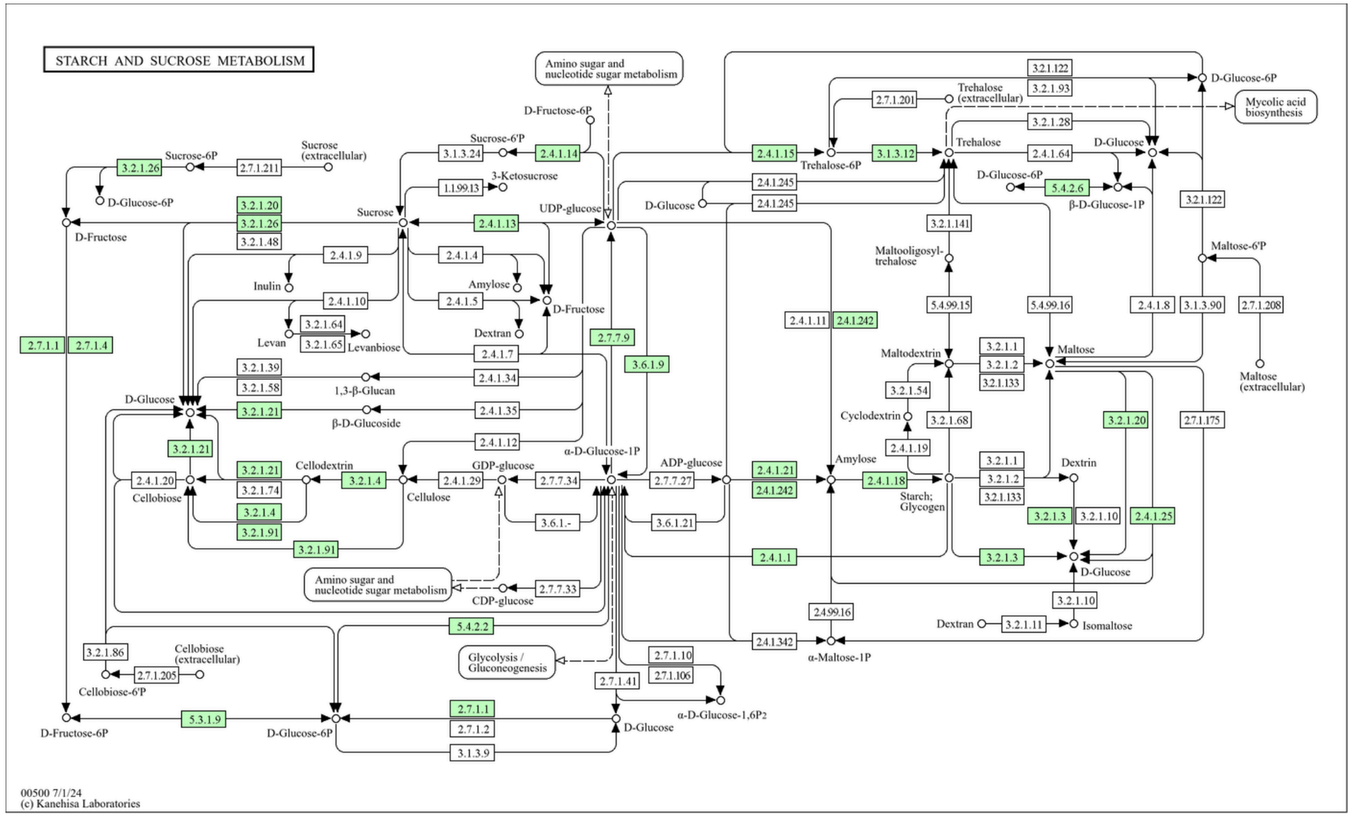
DeepFRI: Predicts protein functions by leveraging sequence features extracted from a protein language model and protein structures.



Pathway completeness

KEGG Pathway Mapper: Maps annotated genes to metabolic pathways using the KEGG database to reconstruct organismal or community metabolism.



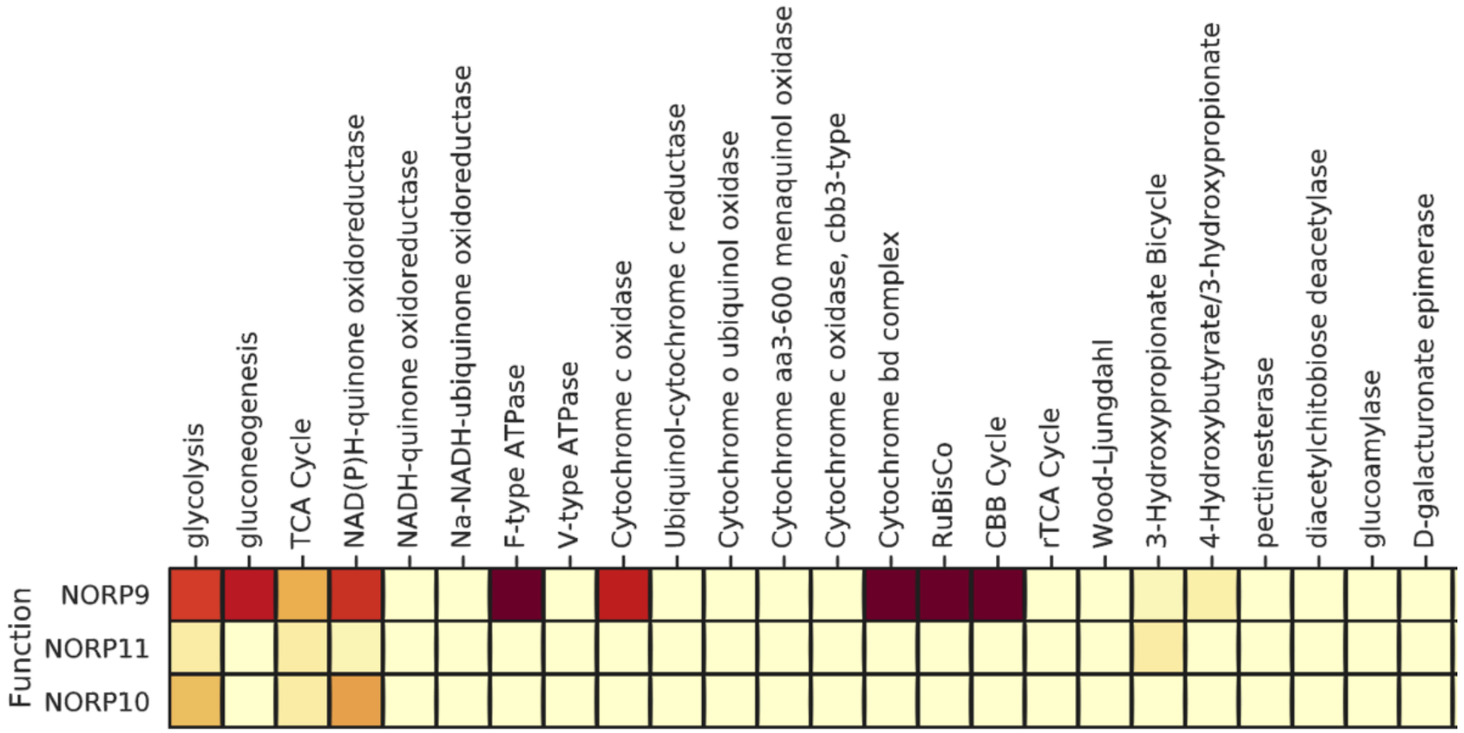


Alternative:

- BioCyc

Pathway completeness

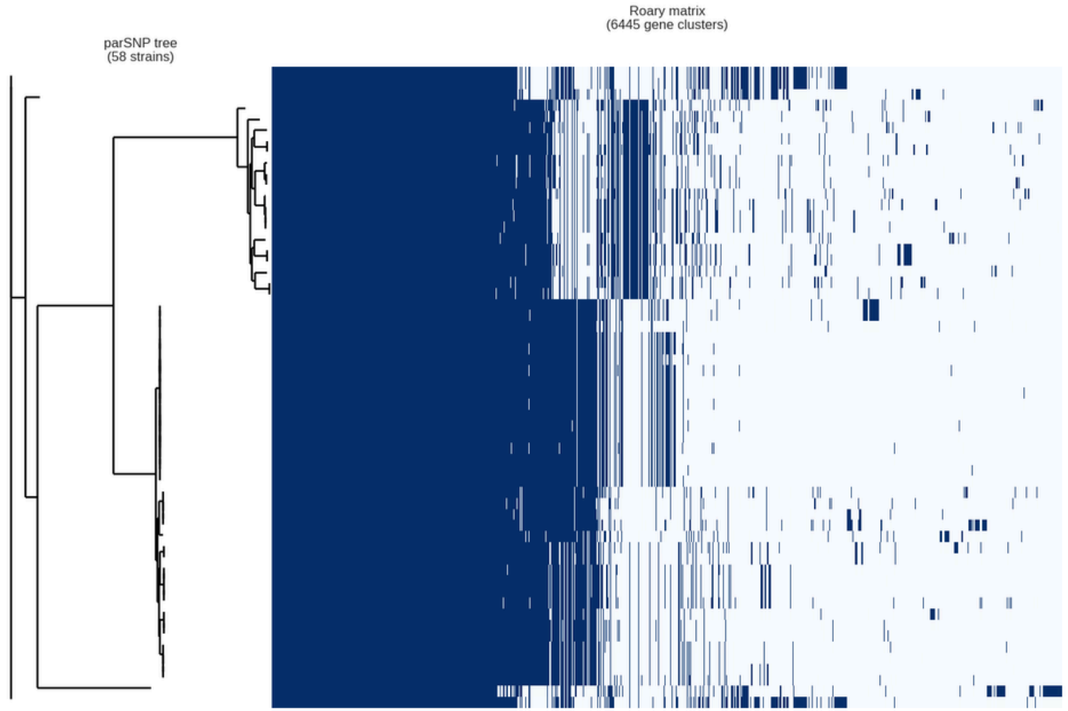
KEGG Decoder: Estimates the completeness of various biogeochemically relevant metabolic pathways.



Not updated.

Biological functions

Roary: Takes annotated assemblies in GFF3 format to calculate the entire set of genes or genetic sequences found within a specific species or population (pangenome).



Methods:

- DNA → Protein
- Clustering (CD-HIT, MCL)
- Graph construction (order of accurrrance)

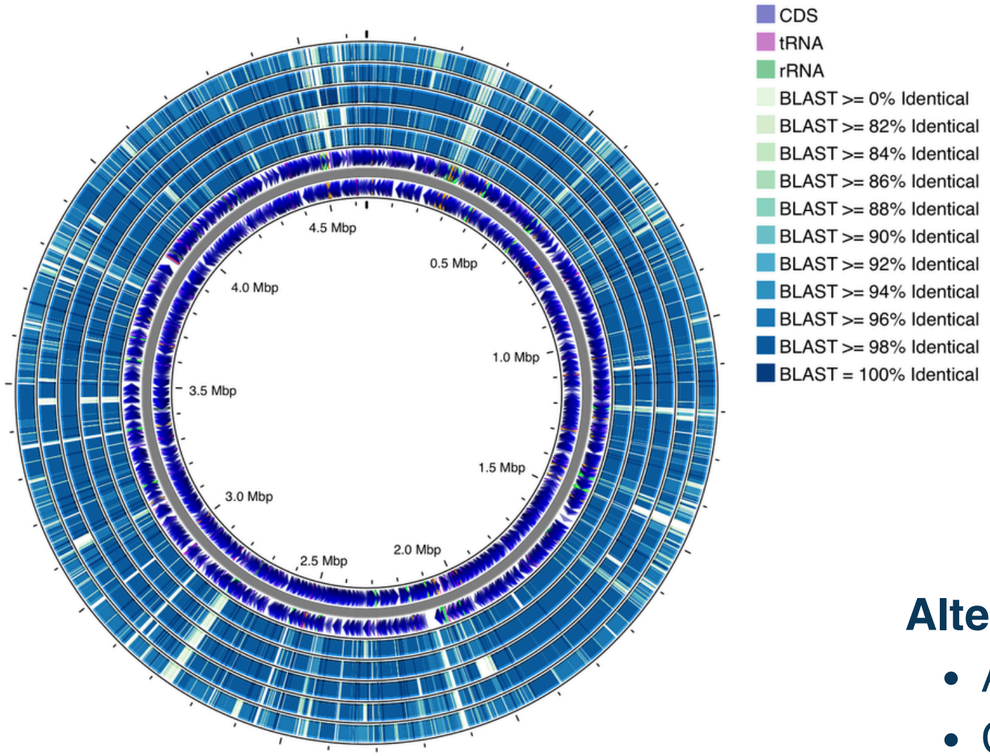
Not updated.

Alternative:

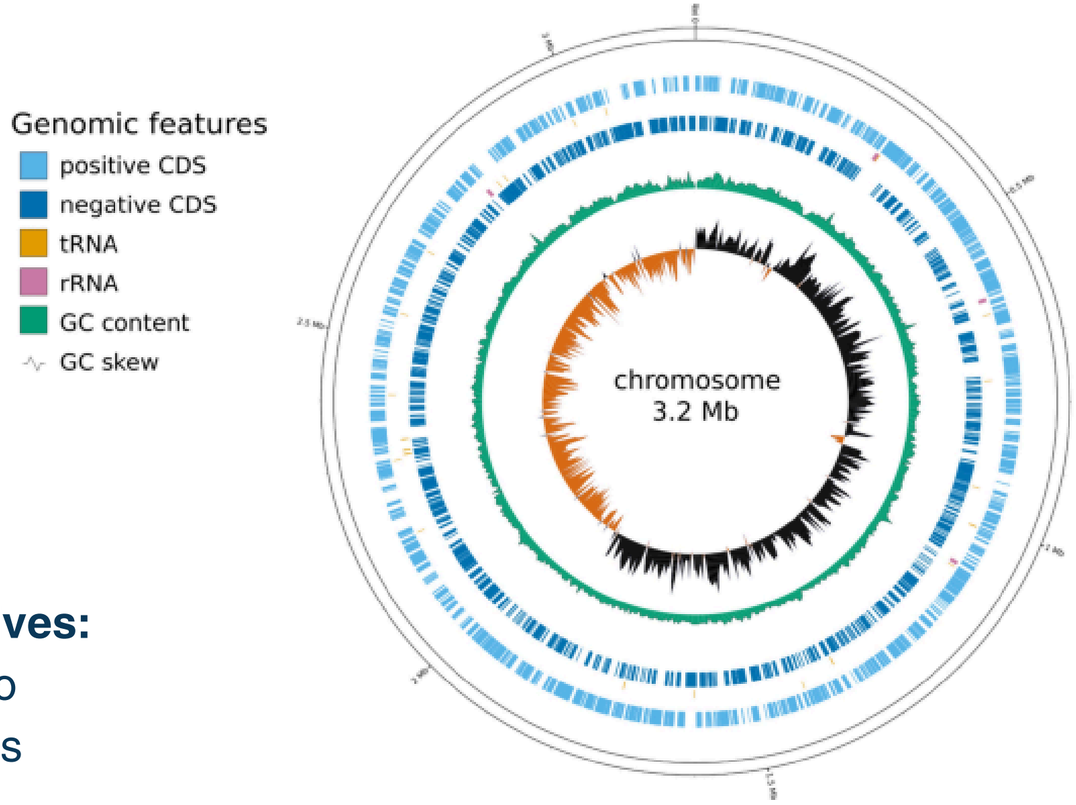
- nf-core/pangenome

Visualization

Proksee: System for genome assembly, annotation and visualization, featuring interactive circular and linear genome maps.



GenoVi: Generates circular genome representations for complete, draft, and multiple bacterial and archaeal genomes.



Alternatives:

- Anvi'o
- Circos
- iTOL

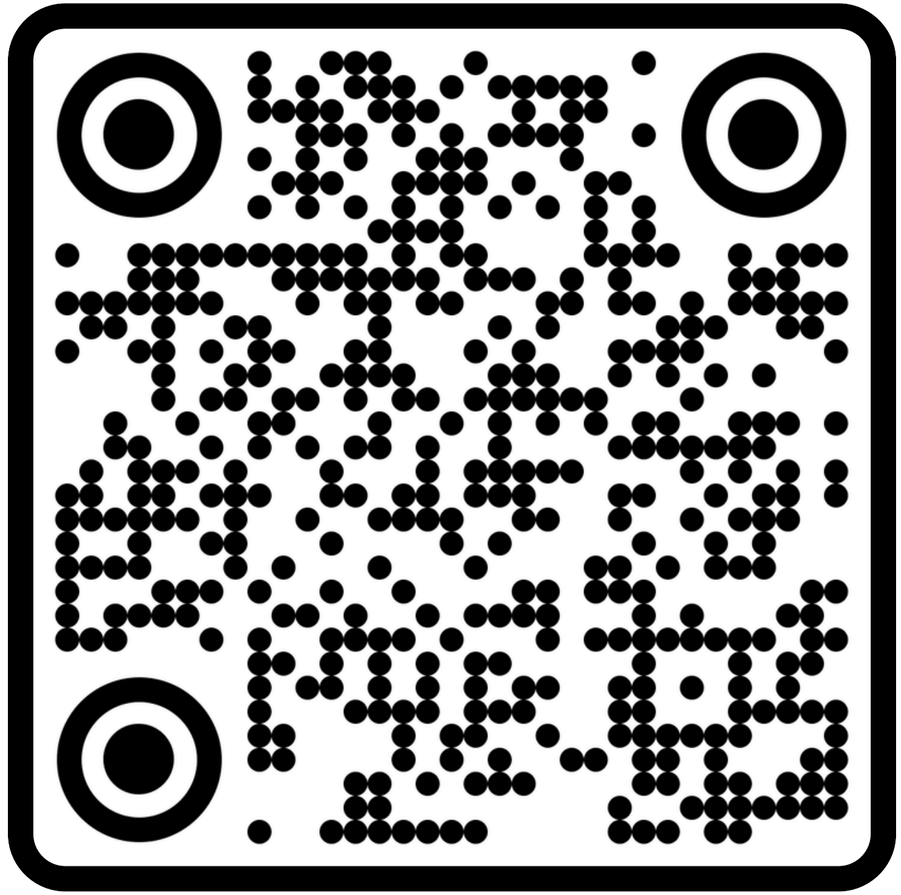
Challenges

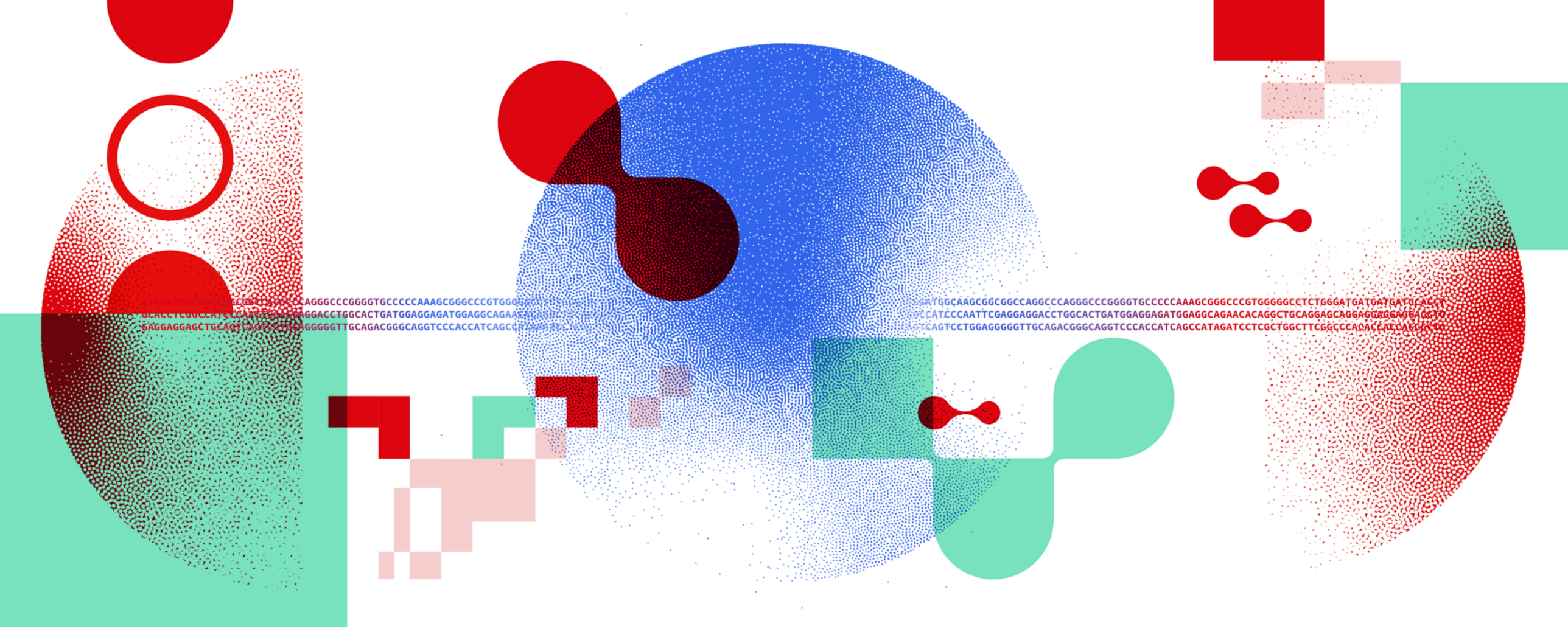
- Incomplete genomes (fragmented, missing genes).
- Contamination and misassembly issues.
- Lack of close reference genomes (novelty of many MAGs).
- Biases in existing functional databases.

Summary

- MAG functional annotation is a multi-step process that requires several tools and databases.
- Combination of tools with different scope and functionalities can expand the knowledge about ecological functions a MAG can perform.
- Automated pipelines are interesting approches, although they may be slow, web-server dependent or time-consuming during debugging.

Practicals





Thank you

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